Map Design

Lecture #11 | GEOG 510
GIS & Spatial Analysis in Public Health

Varun Goel

Overview

- Decisions, decisions, decisions
- Symbolization
- Color
- Design principles
- Maps vs figures
- In-class mapping exercise!

Mapping in a GIS

- Mapping for others
  - Main purpose is to communicate geographic information
    - Focus on design/presentation in order to create an output that effectively communicates the information

Decisions in Map Creation

- Information to be presented
  - Selection, generalization, simplification, exaggeration
- Symbolization of information
  - Selecting symbols based on thematic content, including color
- Design and layout of map display
  - Creating an aesthetically pleasing display

Selective Content

- Selecting features and attributes to include in display
  - Determined by the specific objective of the thematic map
- Choosing the appropriate extent
  - What geographic region is most important

Symbolization

- Symbols are chosen to portray information selectively
- Basic spatial features serve as surrogates for real world objects or events
  - Points (e.g. trees), lines (e.g., roads), and areas (e.g., counties)
- Attribute values can be portrayed by varying several symbol characteristics
  - i.e., what?, how much?, how many?, how big?, etc.

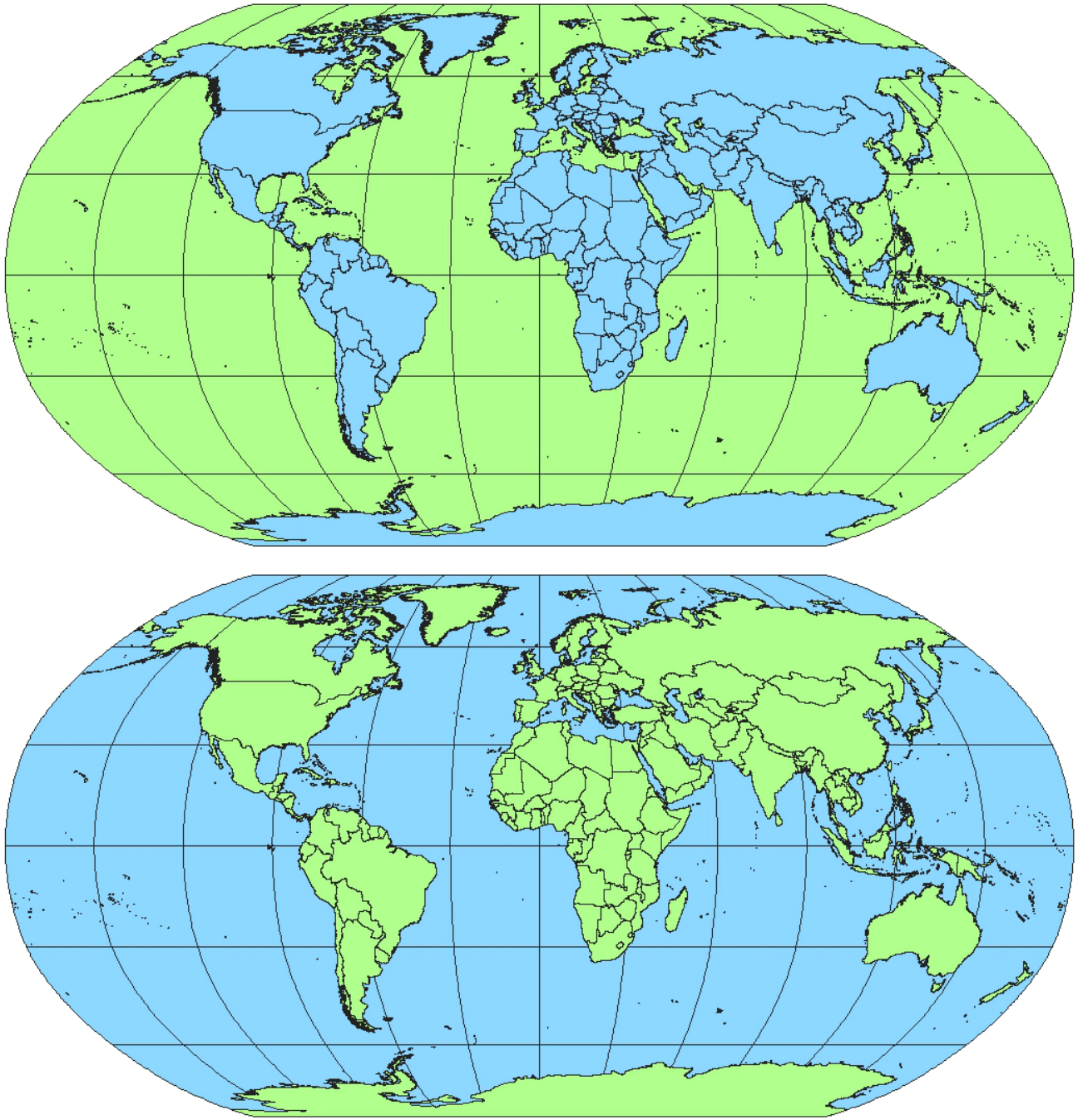
Symbol Characteristics

Feature Type	Visual Variable		
	Shape	Orientation	Color Hue
Point	Spring House Tower	Live Tree Dead Tree	Live Tree Dead Tree
Line	National Border Trail Section Line	Asphalt Road Concrete Road	National Border State Border
Area	Gravel Sand	Orchard Field Crop	Land Water

Feature Type	Visual Variable			
	Size	Pattern Texture	Color Lightness	Color Saturation
Point	Small → Large: Low → High	Coarse → Fine: Low → High	Light → Dark: Low → High	Pale → Intense: Low → High
Line	Low, Medium, High	Low, Medium, High	Low, Medium, High	Low, Medium, High
Area	High, Medium, Low	High, Medium, Low	High, Medium, Low	High, Medium, Low

Color Conventions

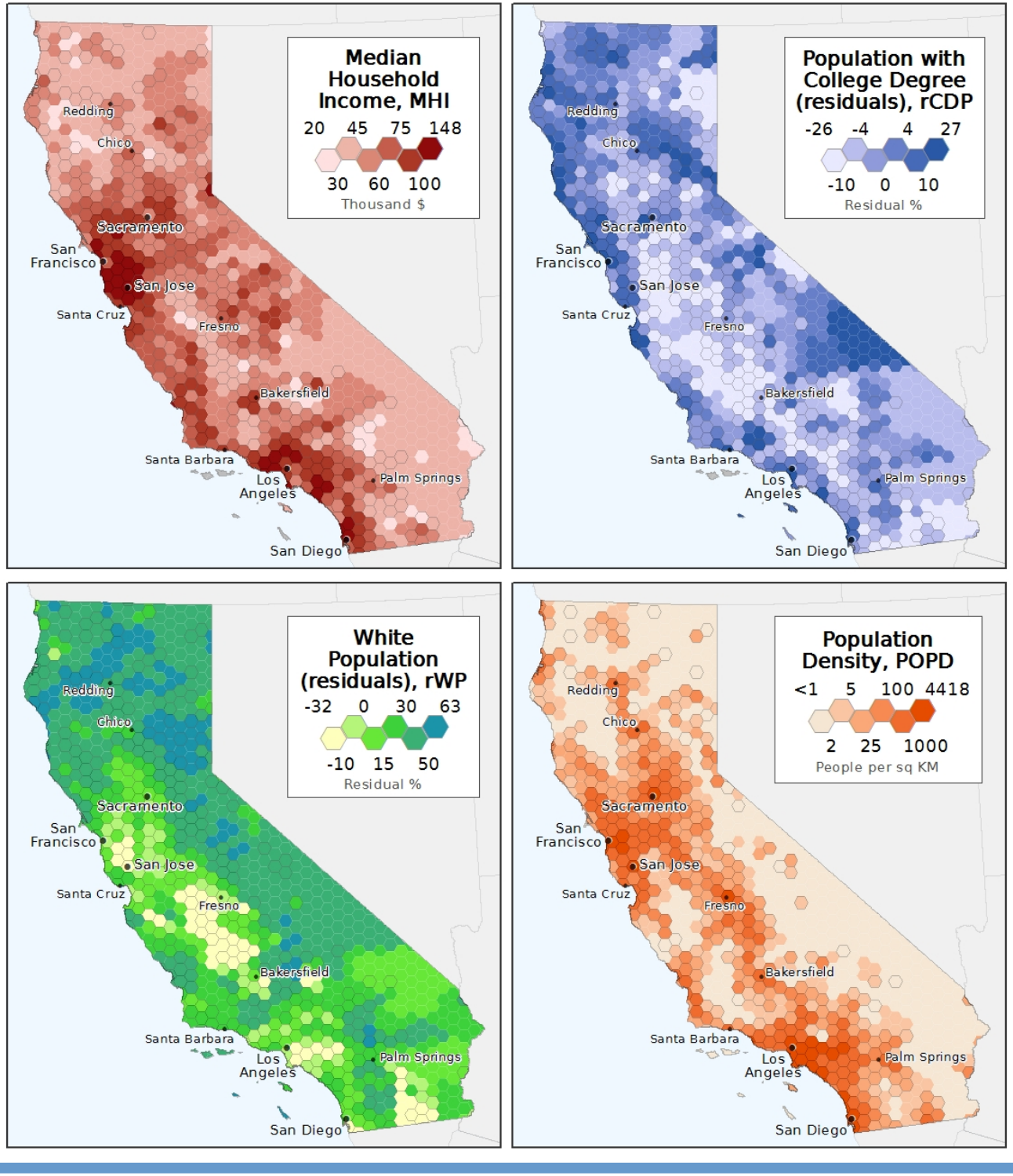
- Specific colors work well for specific physical features
  - e.g., blue for water, green for vegetation, brown for ground or soil
- Specific colors also work well for representing values
  - e.g., red for negative values and blue or green for positive values

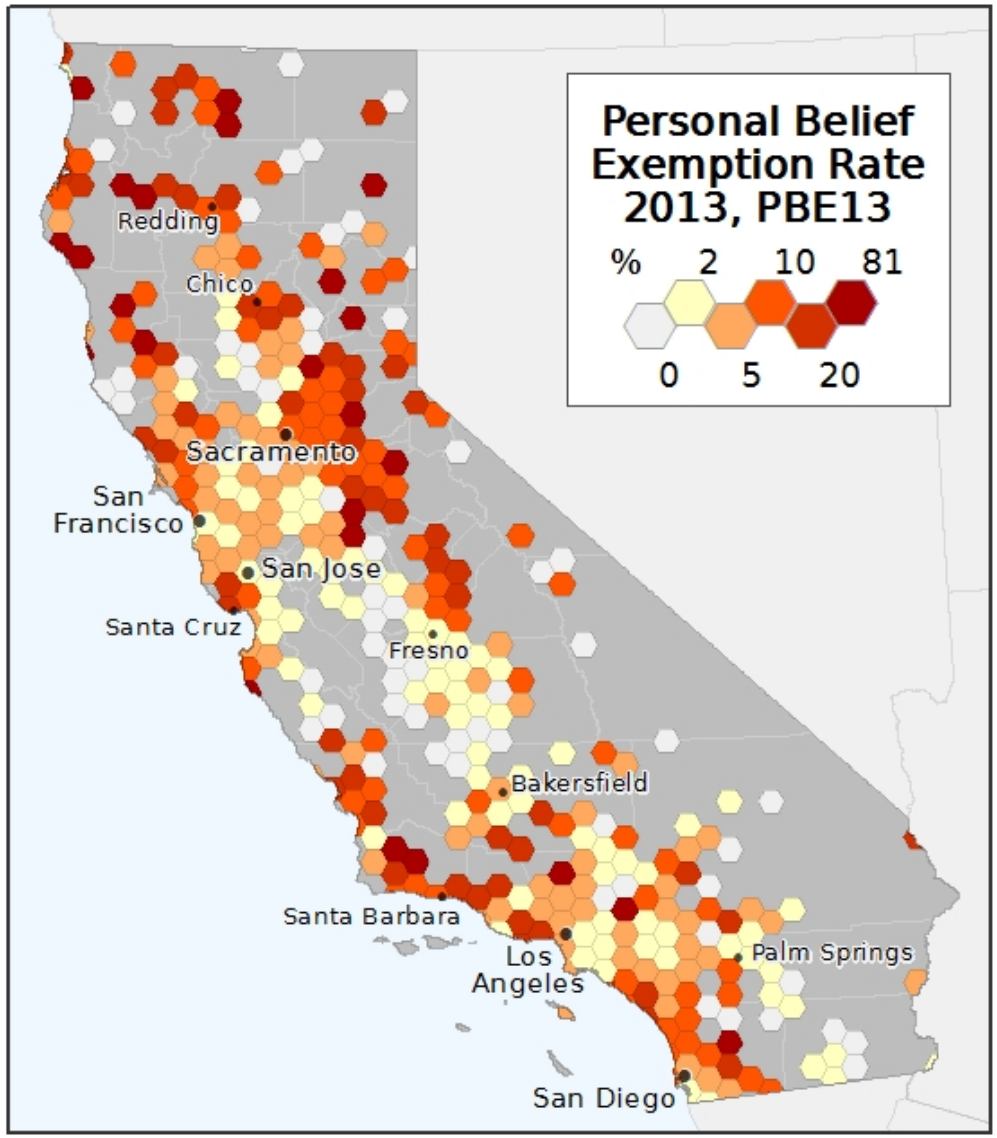


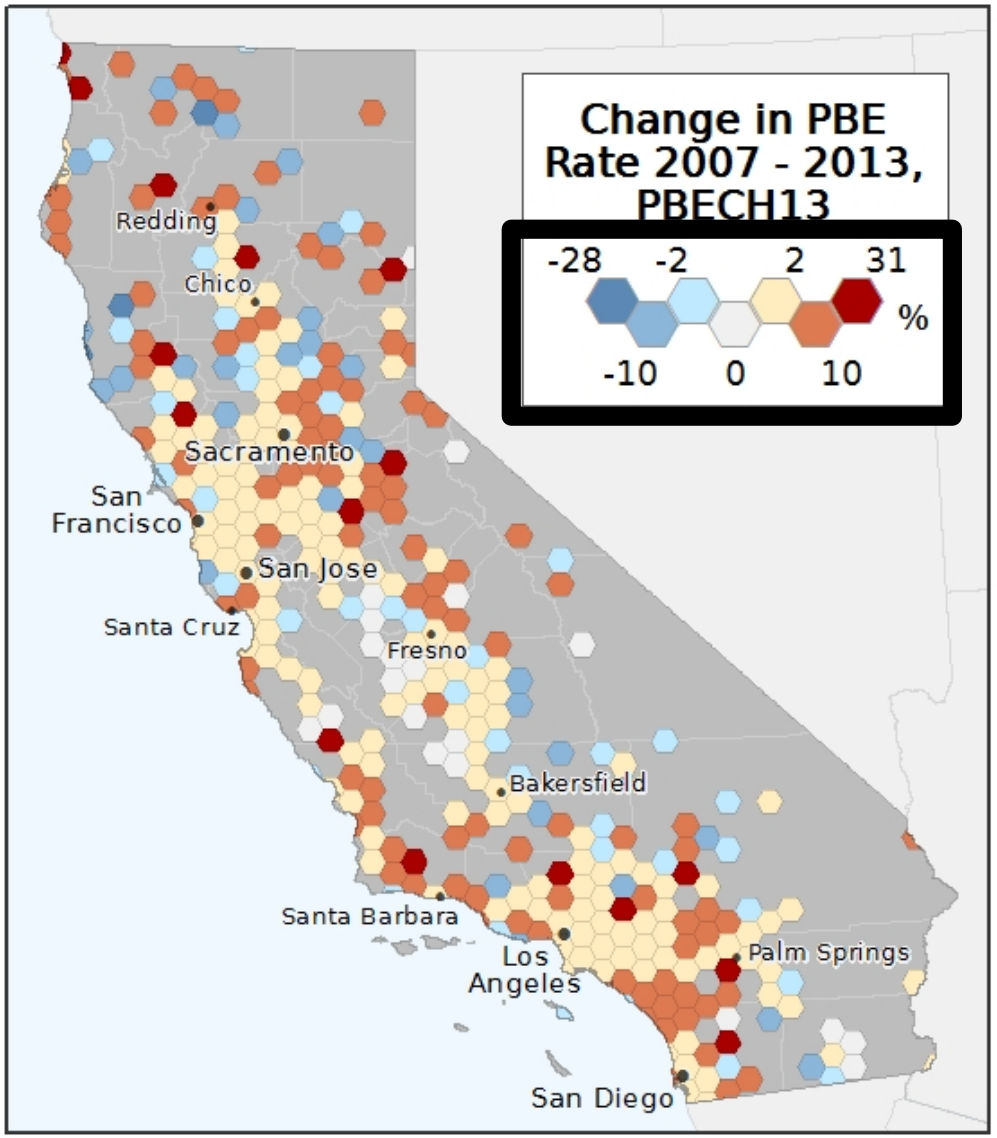
Colors and Quantity

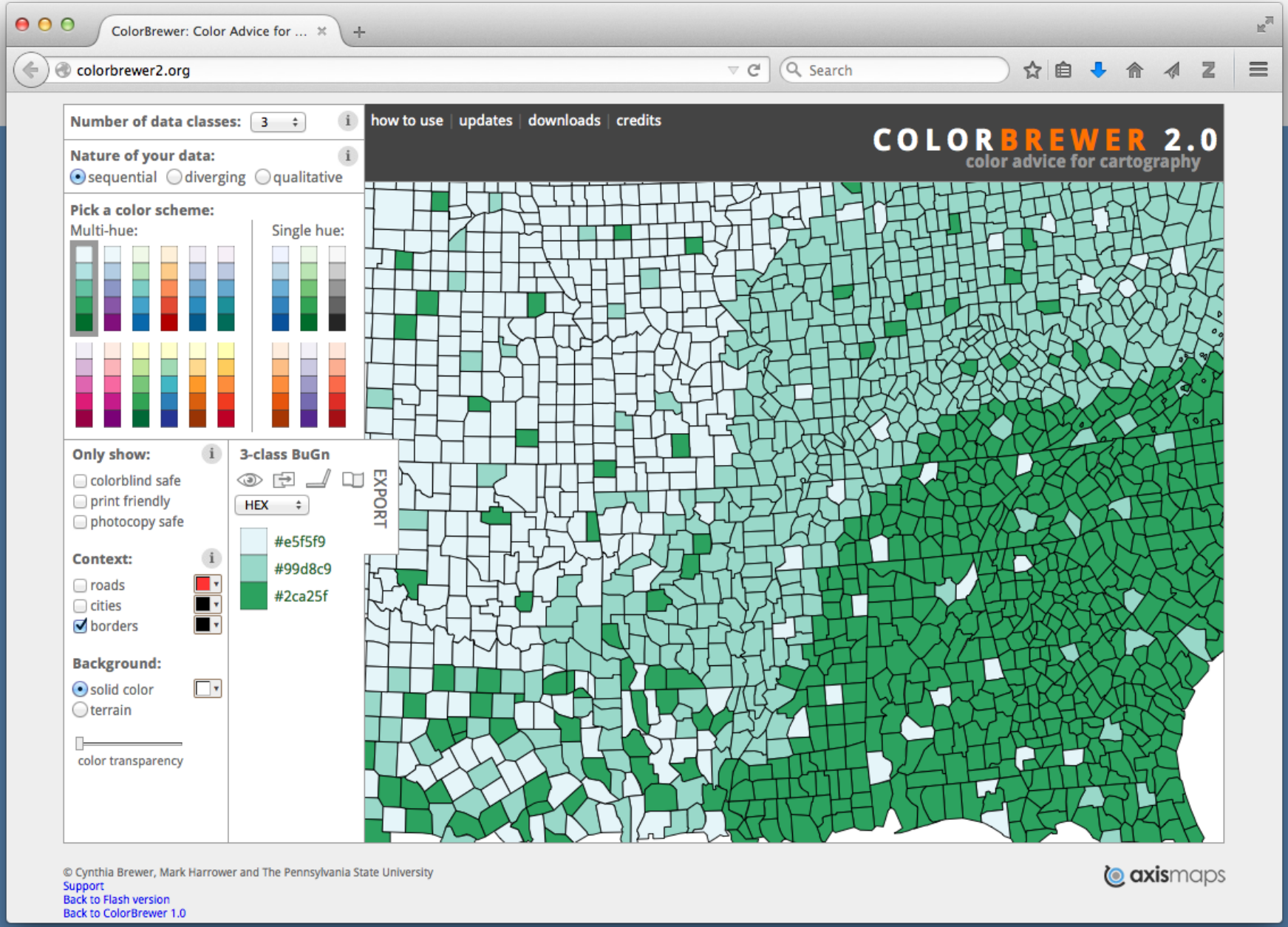
- Color Ramps (value sequences)
  - Most useful for ordinal, interval/ratio variables
  - Use diverging colors, if the variable range has a "true" center
- Blended hue sequences that suggest quantity
  - Spectrum of cool to warm colors
  - Partial spectrum or blending of hues
  - Elevation sequence (blue to green to brown to white)




Median Household Income, MHI
20 45 75 148
30 60 100
Thousand $
Population with College Degree (residuals), rCDP
-26 -4 4 27
-10 0 10
Residual %
White Population (residuals), rWP
-32 0 30 63
-10 15 50
Residual %
Population Density, POPD
<1 5 100 4418
2 25 1000
People per sq KM
Redding
Chico
Sacramento
San Francisco
San Jose
Santa Cruz
Fresno
Bakersfield
Santa Barbara
Los Angeles
Palm Springs
San Diego


Personal Belief Exemption Rate 2013, PBE13
% 2 10 81
0 5 20
Redding
Chico
Sacramento
San Francisco
San Jose
Santa Cruz
Fresno
Bakersfield
Santa Barbara
Los Angeles
Palm Springs
San Diego


Change in PBE Rate 2007 - 2013, PBECH13
-28 -2 2 31
%
-10 0 10
Redding
Chico
Sacramento
San Francisco
San Jose
Santa Cruz
Fresno
Bakersfield
Santa Barbara
Los Angeles
Palm Springs
San Diego

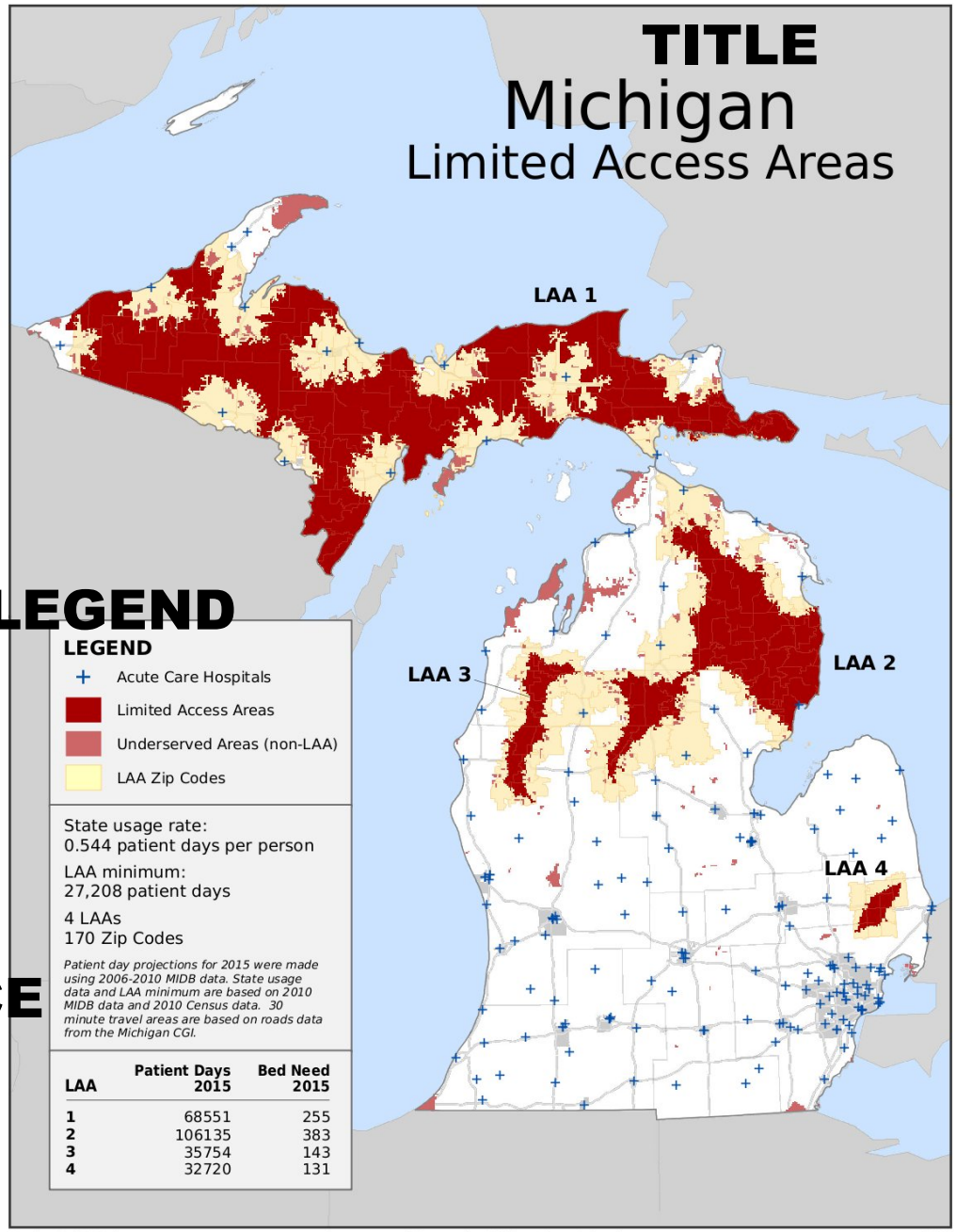


Cynthia Brewer, Pennsylvania State University– http://colorbrewer2.org

Map Elements

- Title: Identifies the map's content
- Legend: Relates symbols to their meaning
- Inset/Locator: Shows relative location of map
- North Arrow: Points “north”
- Information Sources: Allows users to evaluate map suitability
- Scale: Defines the relationship between distance on map and ground

NEATLINE



LEGEND

SOURCE

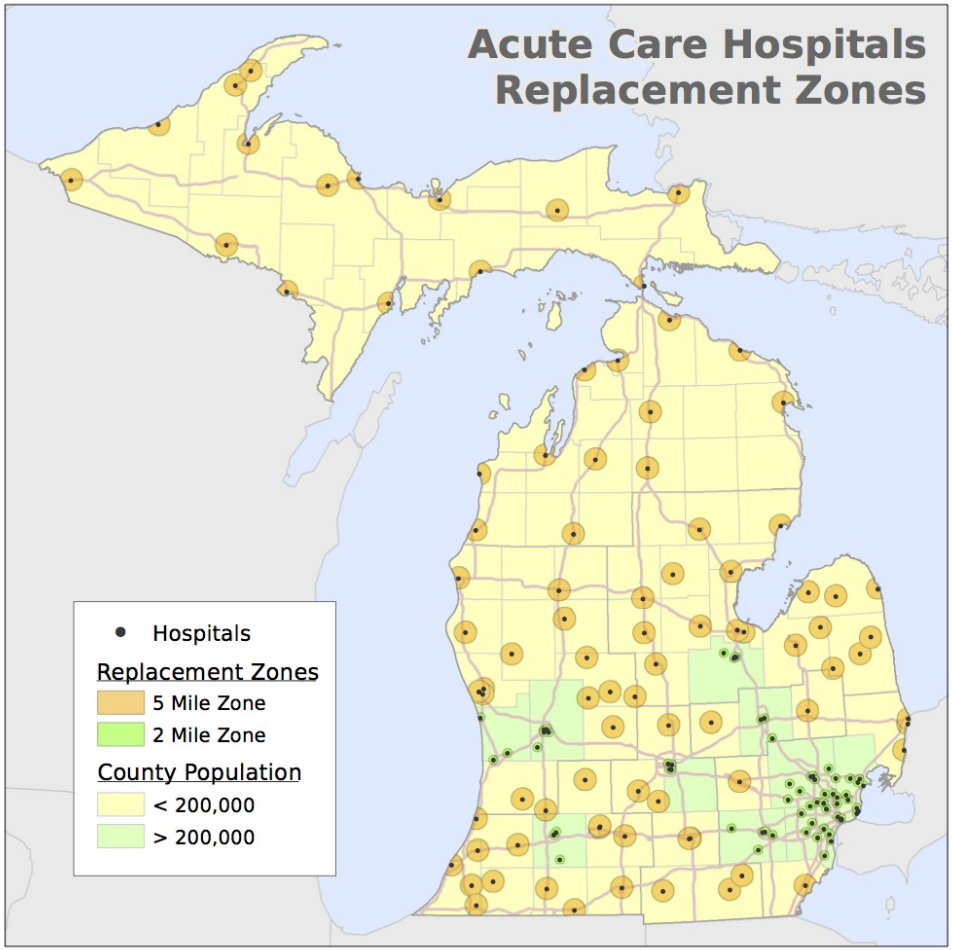
AUTHOR AGENCY DATE

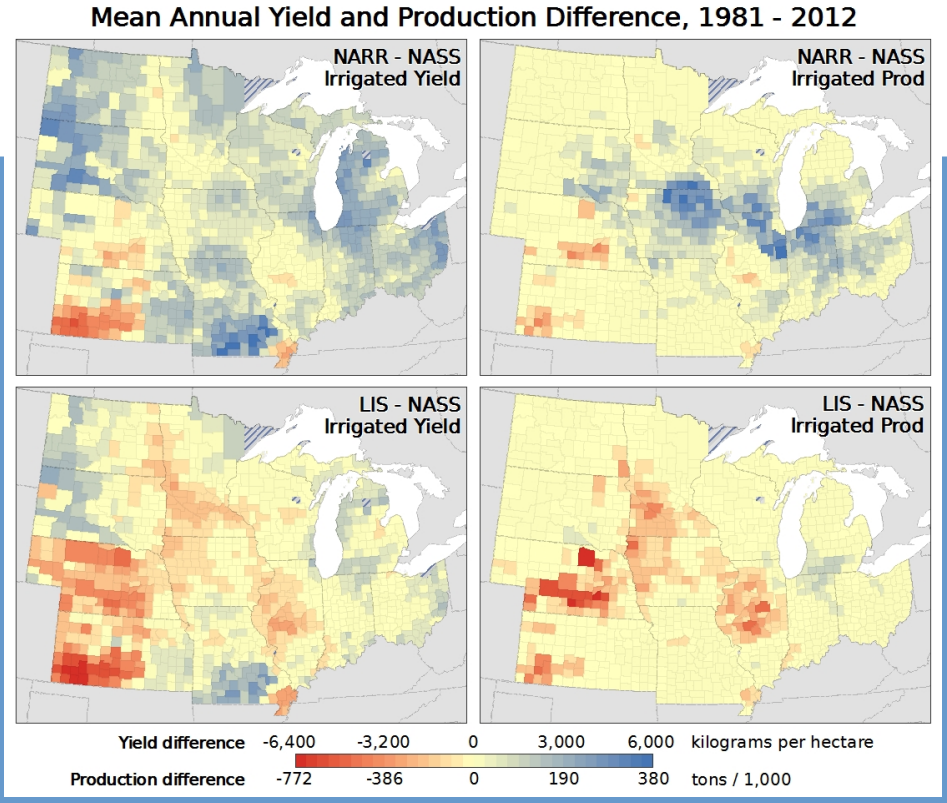
Design Principles

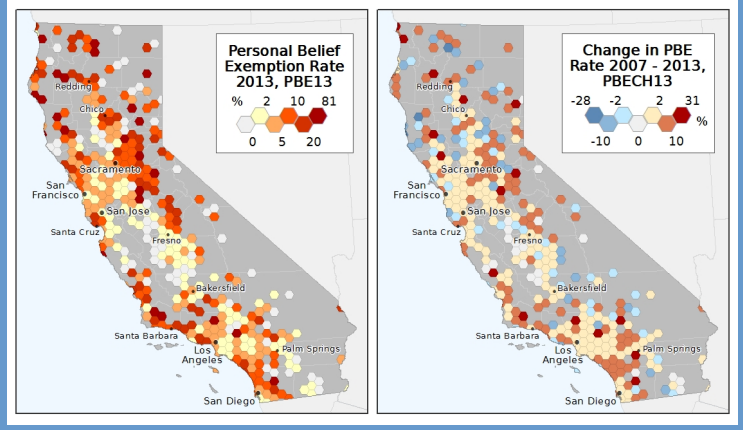
- Balance
- Hierarchy
- Figure-Ground
- Legibility

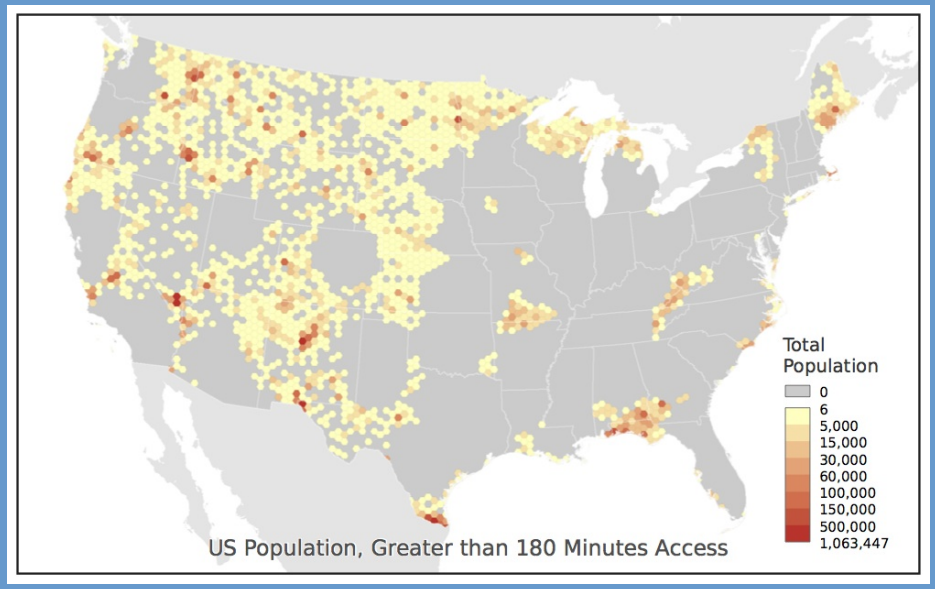
Balance

- Experiment with placement of all map elements on the page
- Be sure that the map itself is as large as possible
- No single layout is correct or incorrect, but properly balanced layouts minimize blank space while making efficient use of space to convey information
- Prioritize information by size


Acute Care Hospitals
Replacement Zones
Hospitals
Replacement Zones
5 Mile Zone
2 Mile Zone
County Population
< 200,000
> 200,000

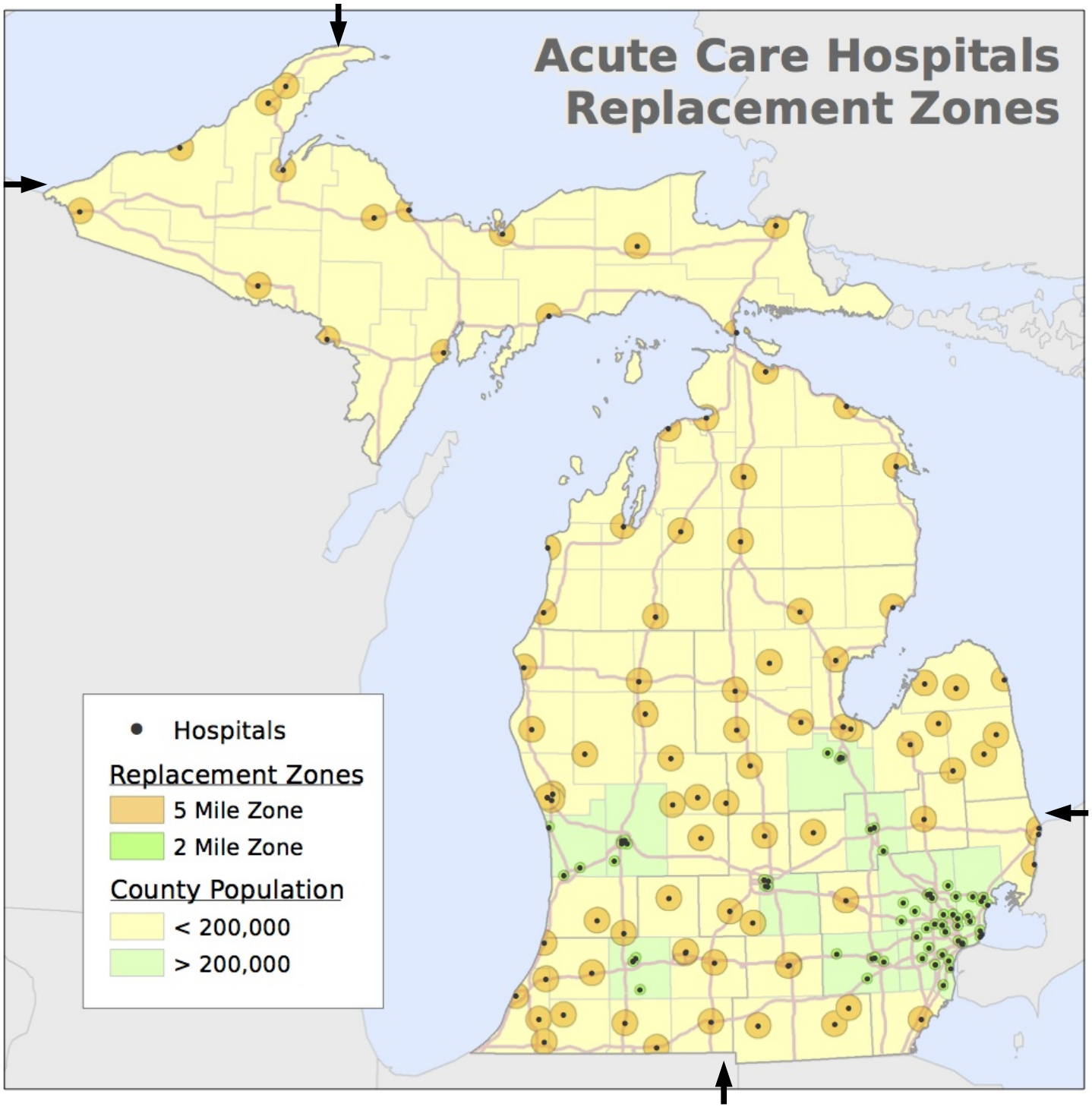

Mean Annual Yield and Production Difference, 1981 - 2012
NARR - NASS
Irrigated Yield
NARR - NASS
Irrigated Prod
LIS - NASS
Irrigated Yield
LIS - NASS
Irrigated Prod
Yield difference -6,400 -3,200 0 3,000 6,000 kilograms per hectare
Production difference -772 -386 0 190 380 tons / 1,000

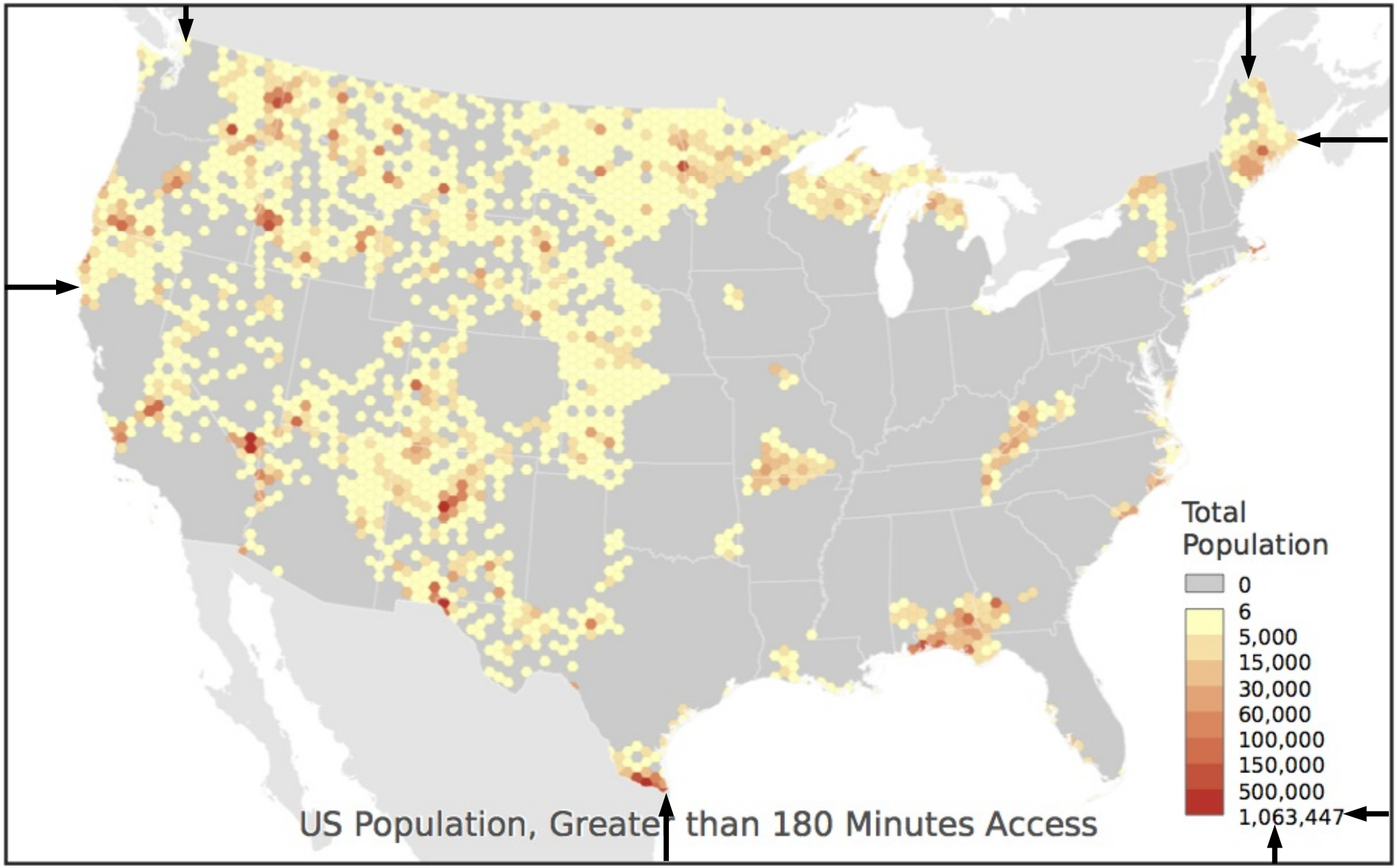

Personal Belief
Exemption Rate
2013, PBE13
% 2 10 81
0 5 20
Change in PBE
Rate 2007 - 2013,
PBECH13
-28 -2 2 31 %
-10 0 10
Redding
Chico
Sacramento
San Francisco
San Jose
Santa Cruz
Fresno
Bakersfield
Santa Barbara
Los Angeles
Palm Springs
San Diego

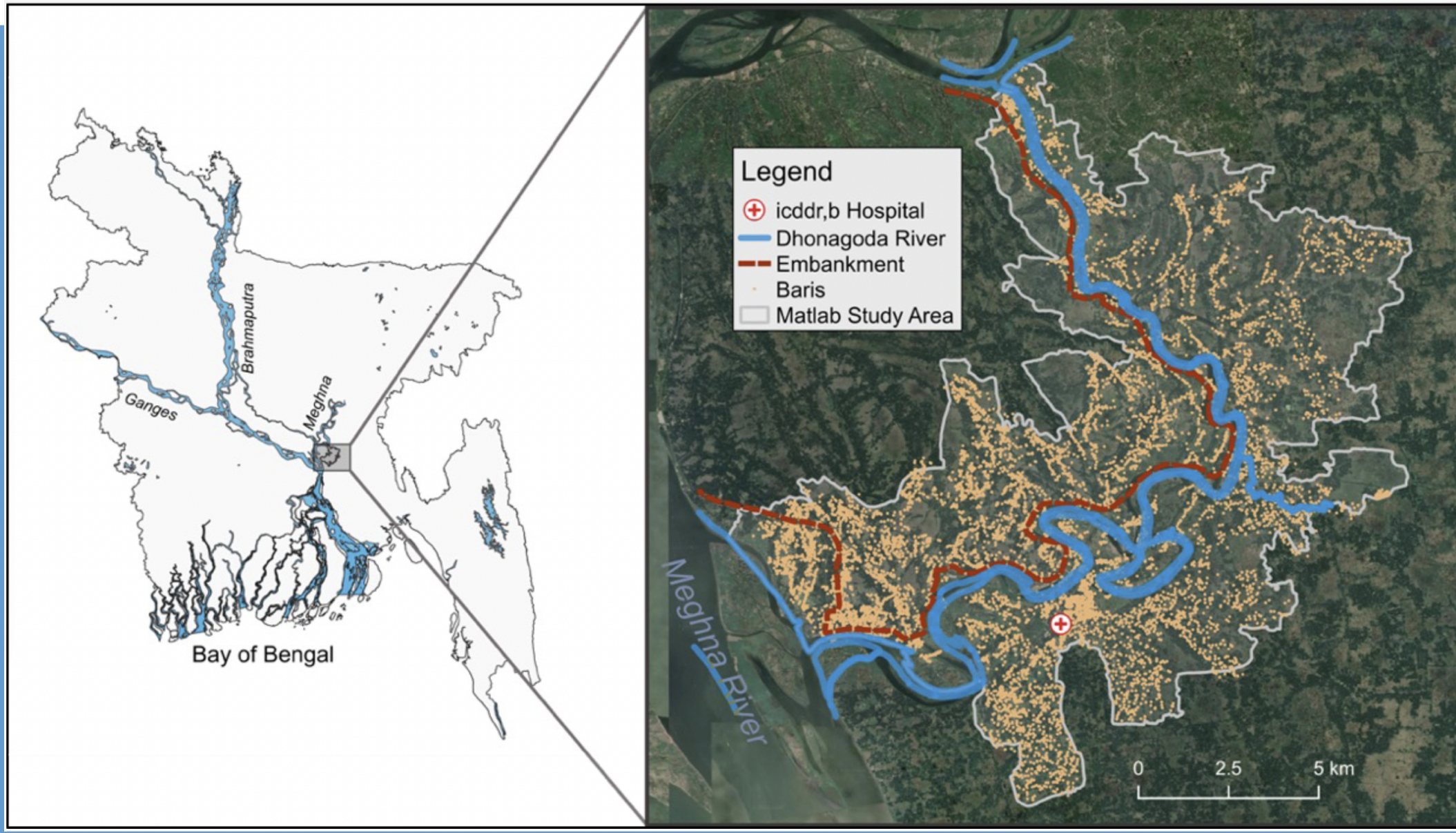

Total
Population
0
6
5,000
15,000
30,000
60,000
100,000
150,000
500,000
1,063,447
US Population, Greater than 180 Minutes Access

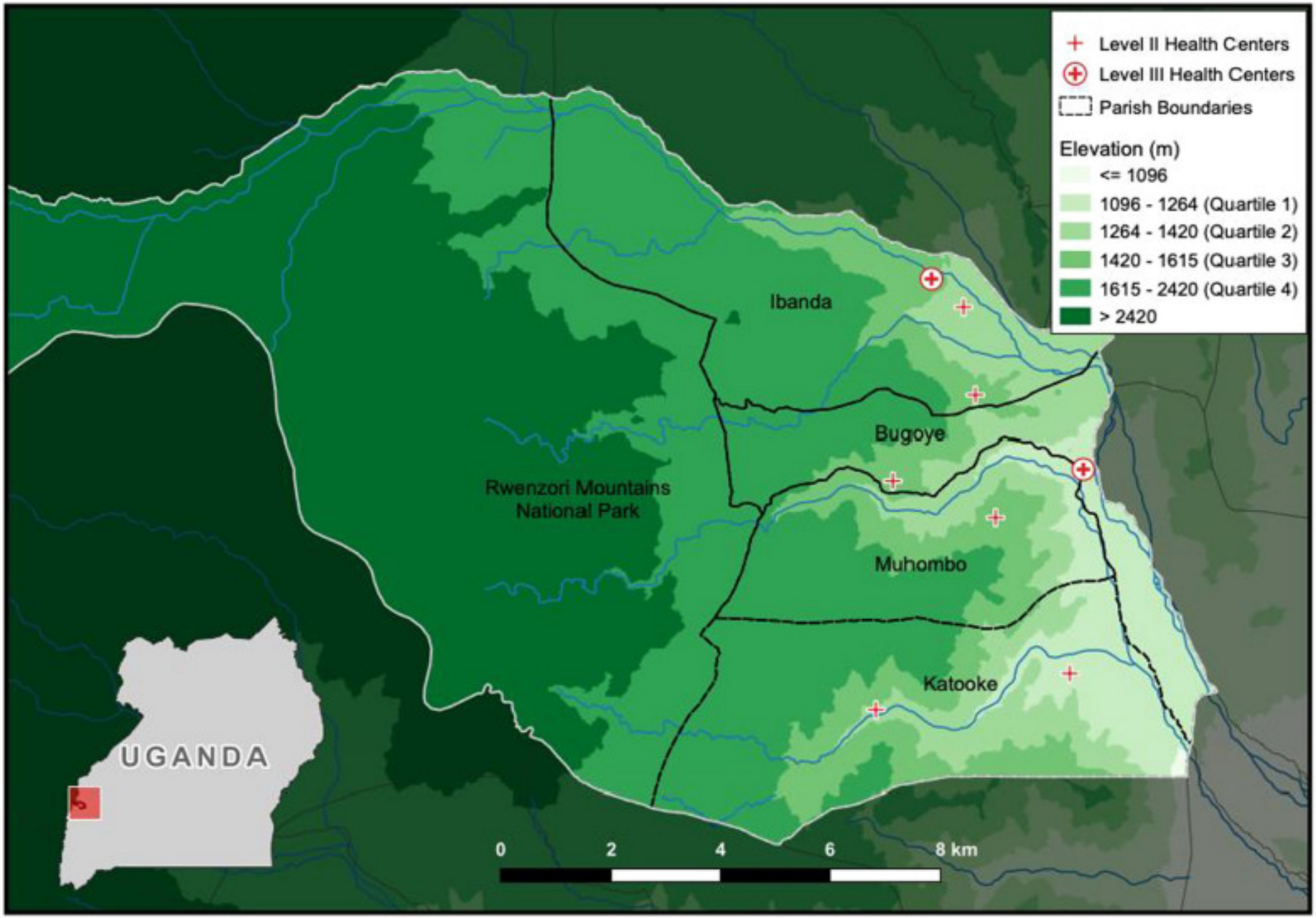
Use combination of Zoom and Pan tools to position the study area or area of interest within the frame

Make study area as large as possible within the neatline bounds, allowing for other elements




Total
Population
0
6
5,000
15,000
30,000
60,000
100,000
150,000
500,000
1,063,447
US Population, Greater than 180 Minutes Access


Ganges
Brahmaputra
Meghna
Bay of Bengal
Legend
icddr,b Hospital
Dhonagoda River
Embankment
Baris
Matlab Study Area
Meghna River
0
2.5
5 km



Level II and Level III health centres.

Hierarchy

- Relative importance of information should be clear
- Placement can be one way to indicate importance of information
- Roughly, top left tends to receive more attention from map readers

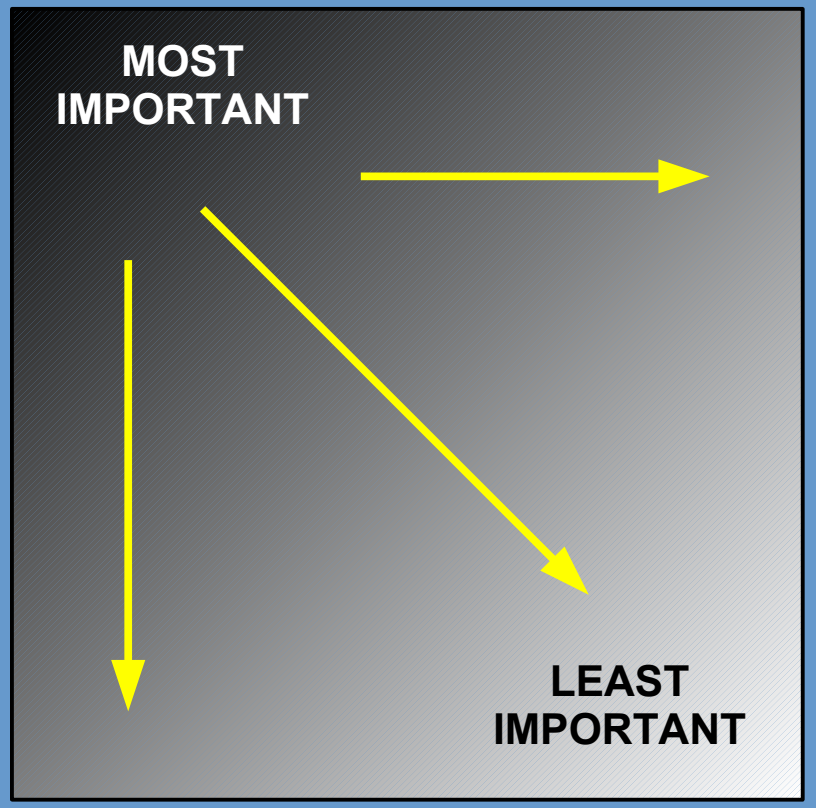
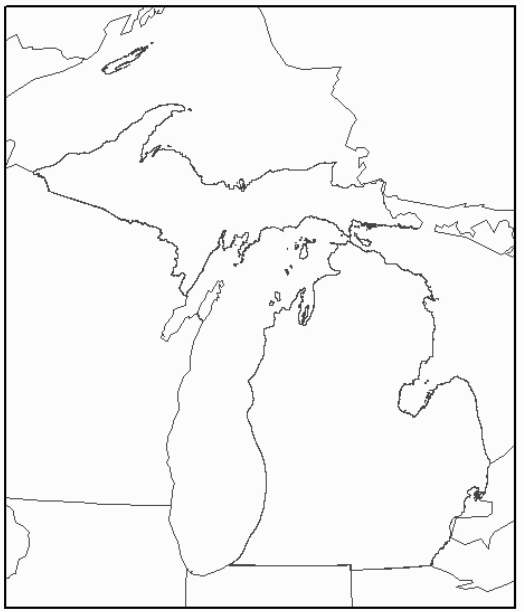


Figure-Ground

- Be clear about what is the information to be highlighted and what is the background
- Use contrast and color to highlight the important information against a less-important background





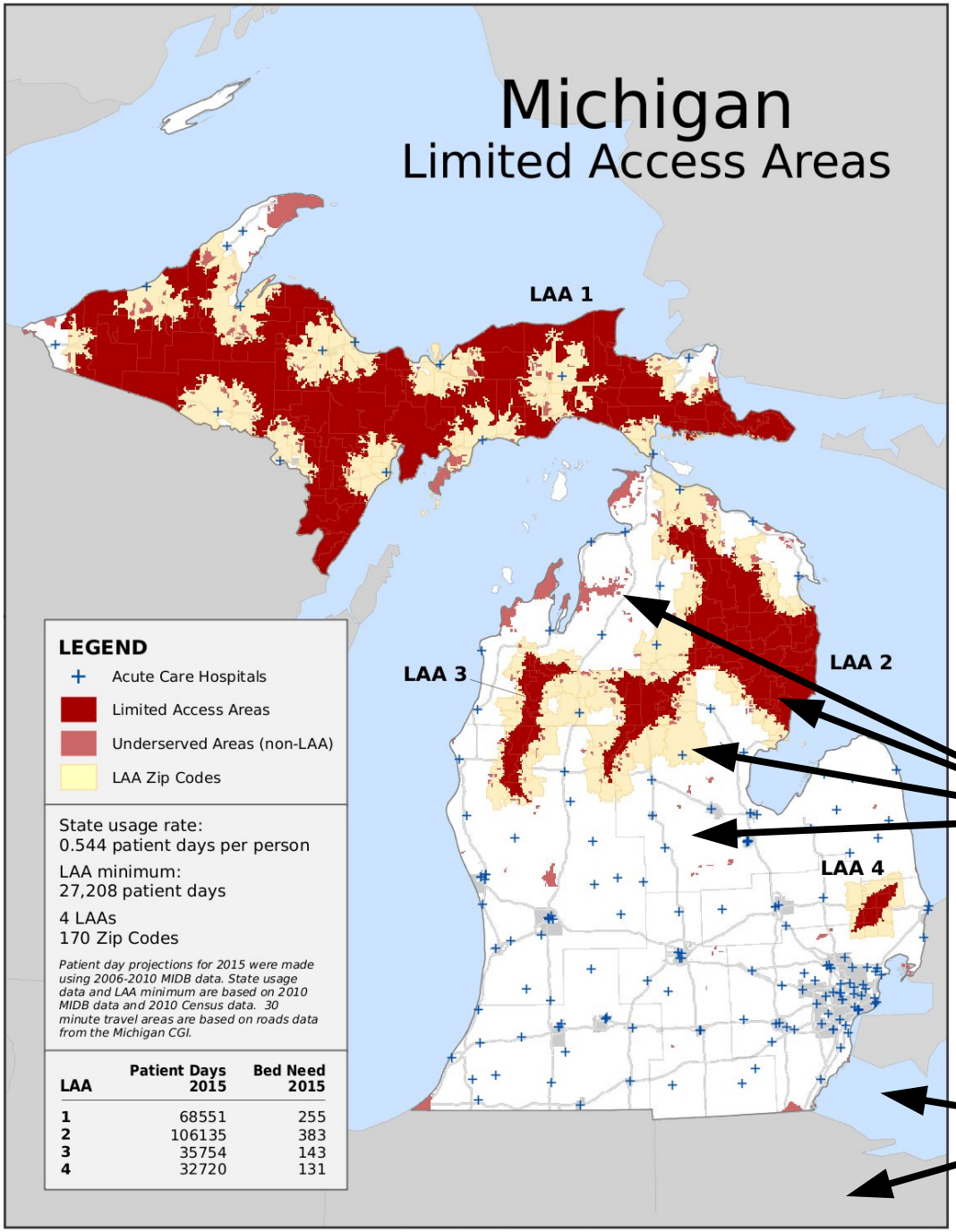
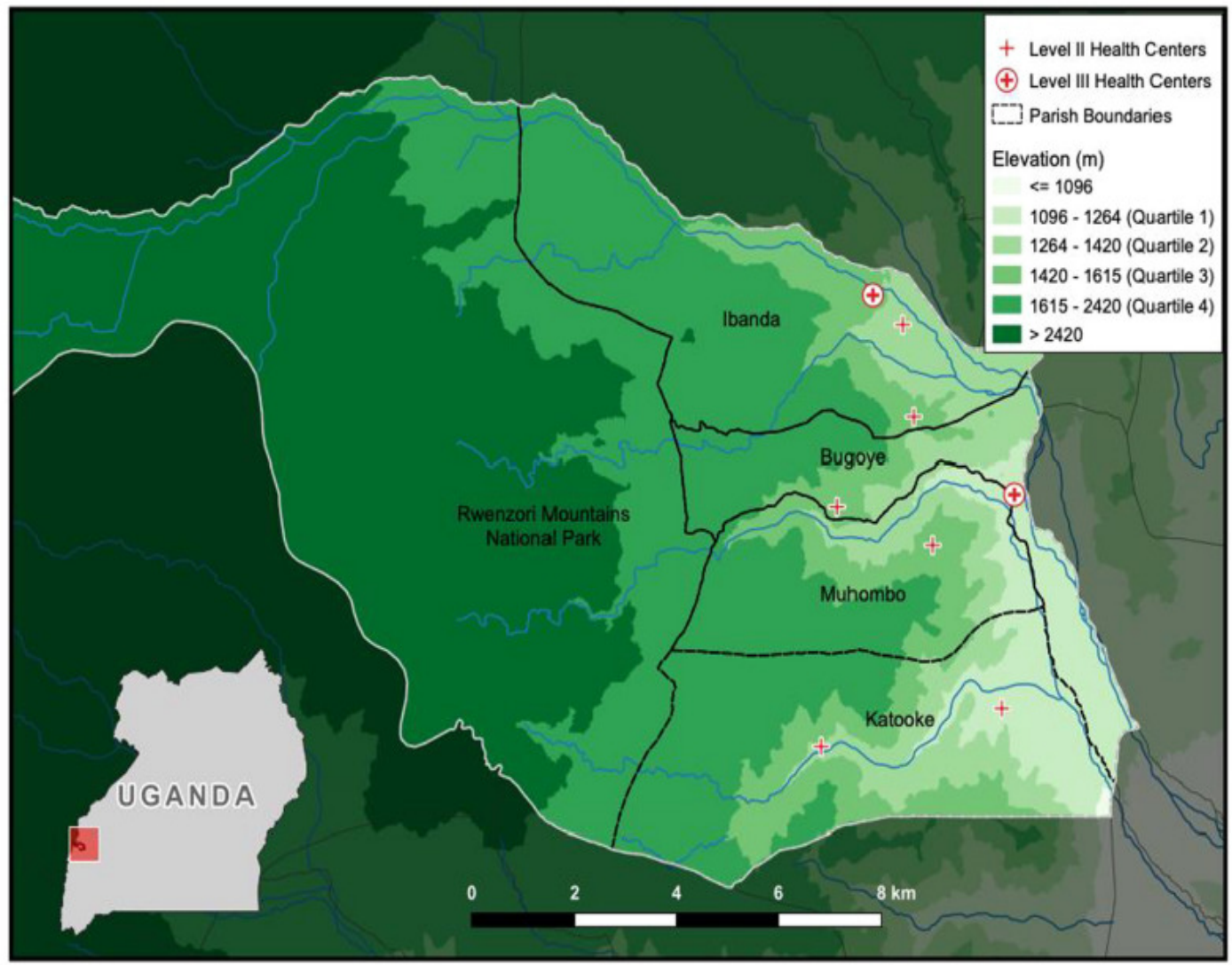


Figure colors

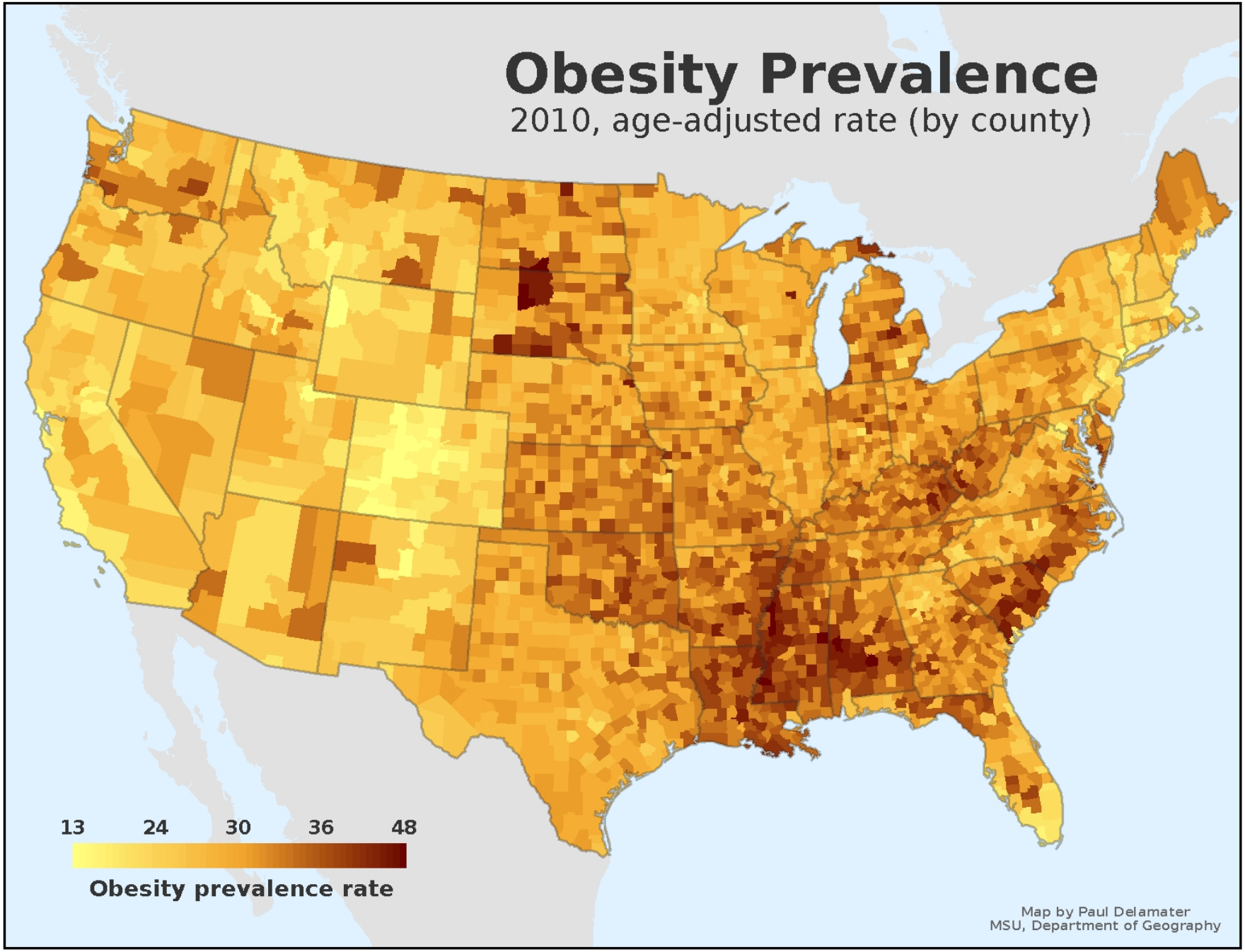
Ground colors

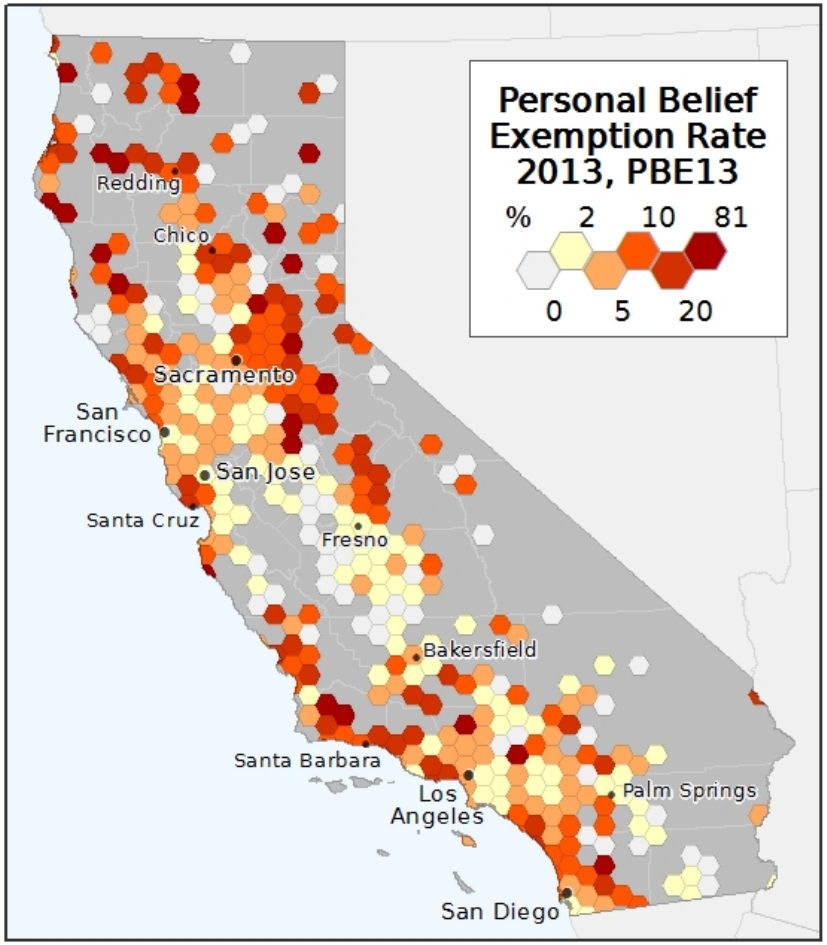


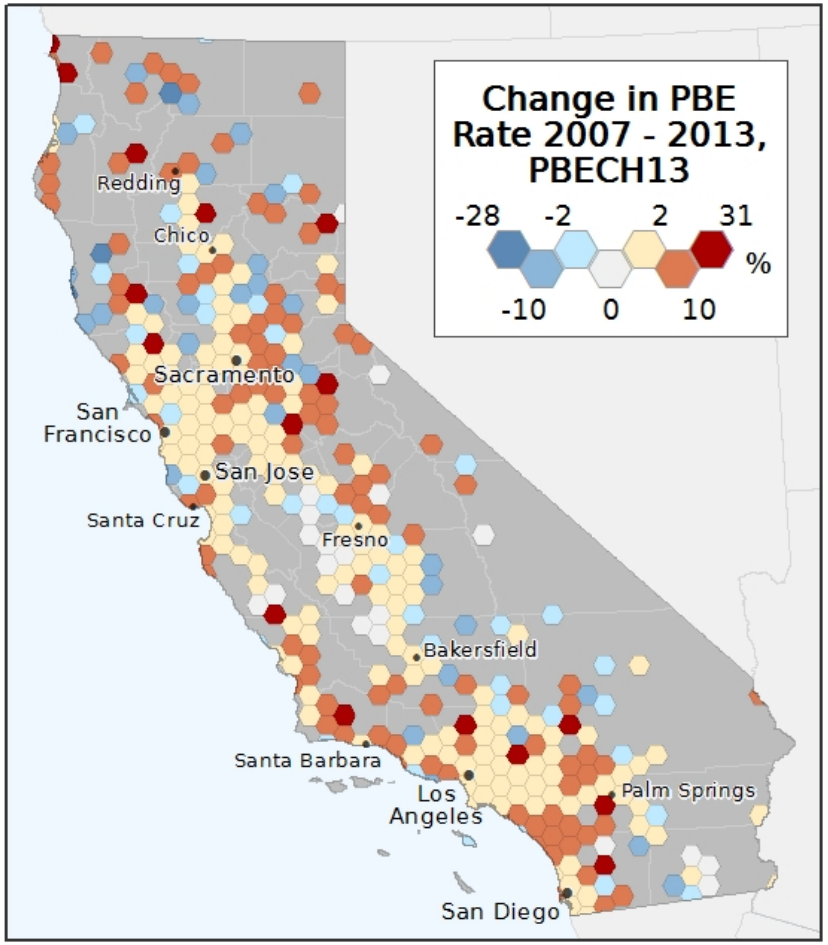
Level II and Level III health centres.

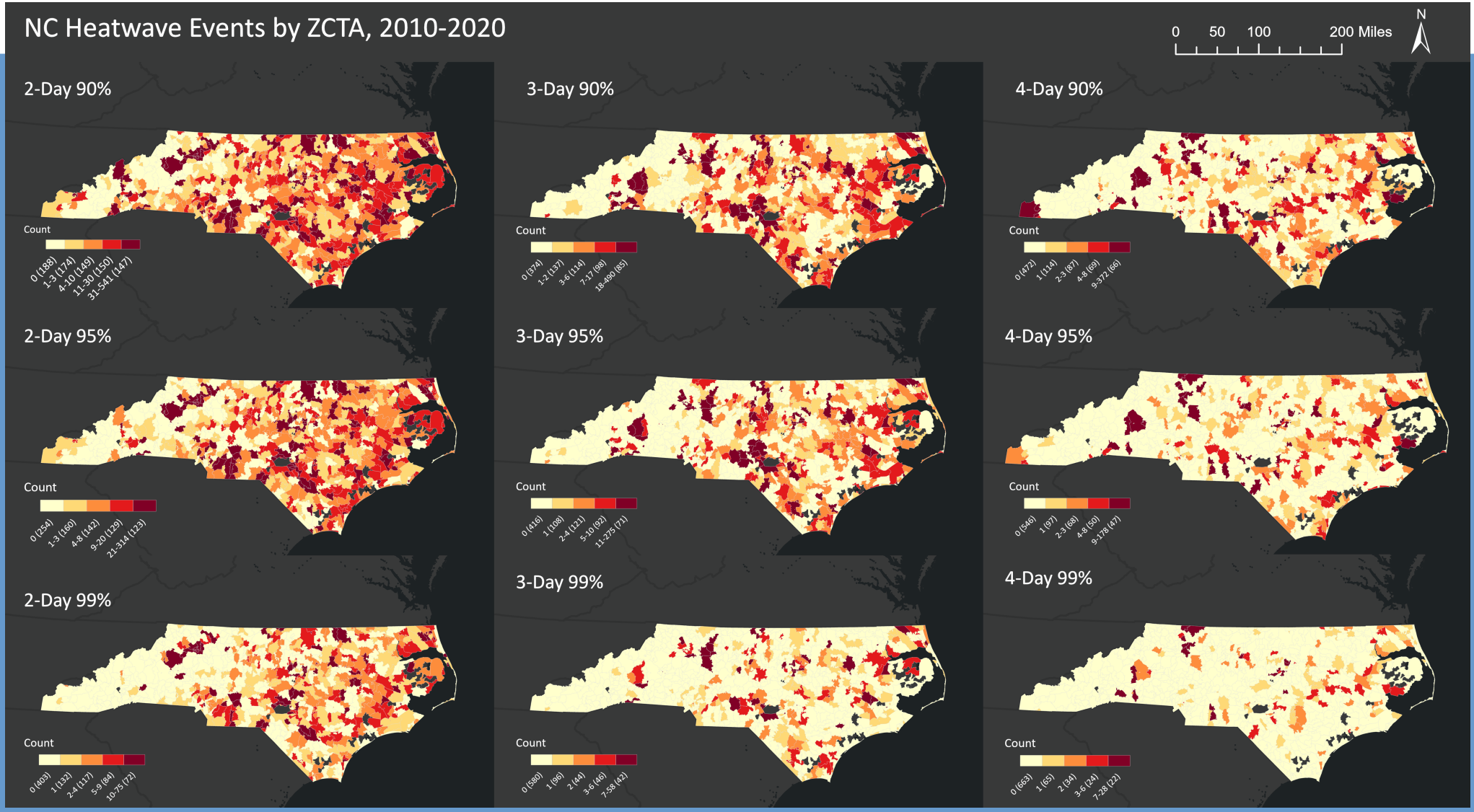
e colors

nd colors


Obesity Prevalence
2010, age-adjusted rate (by county)
13
24
30
36
48
Obesity prevalence rate
Map by Paul Delamater
MSU, Department of Geography


Personal Belief Exemption Rate 2013, PBE13
% 2 10 81
0 5 20
Redding
Chico
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San Francisco
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Change in PBE Rate 2007 - 2013, PBECH13
-28 -2 2 31
%
-10 0 10
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Chico
Sacramento
San Francisco
San Jose
Santa Cruz
Fresno
Bakersfield
Santa Barbara
Los Angeles
Palm Springs
San Diego


NC Heatwave Events by ZCTA, 2010-2020
0 50 100 200 Miles
N
2-Day 90%
Count
0 (188)
1-3 (174)
4-10 (149)
11-30 (150)
31-541 (147)
3-Day 90%
Count
0 (374)
1-2 (137)
3-6 (114)
7-17 (98)
18-490 (65)
4-Day 90%
Count
0 (472)
1 (114)
2-3 (87)
4-8 (69)
9-372 (66)
2-Day 95%
Count
0 (254)
1-3 (160)
4-8 (142)
9-20 (129)
21-314 (123)
3-Day 95%
Count
0 (416)
1 (108)
2-4 (121)
5-10 (92)
11-275 (71)
4-Day 95%
Count
0 (546)
1 (97)
2-3 (68)
4-8 (50)
9-178 (47)
2-Day 99%
Count
0 (469)
1 (132)
2-4 (117)
5-9 (84)
10-75 (72)
3-Day 99%
Count
0 (580)
1 (96)
2 (44)
3-6 (46)
7-58 (42)
4-Day 99%
Count
0 (663)
1 (65)
2 (34)
3-6 (24)
7-28 (22)

Legibility

- Less is more
  - Be sure that everything on the map is necessary and that it communicates as effectively as possible
- Be sure that fonts are not too large or small
  - Use a common font (or only two) throughout
    - Stay away from
      - Comic Sans, Calligraphy/script, Handwriting, **Superbold,** etc.

Figures/Maps

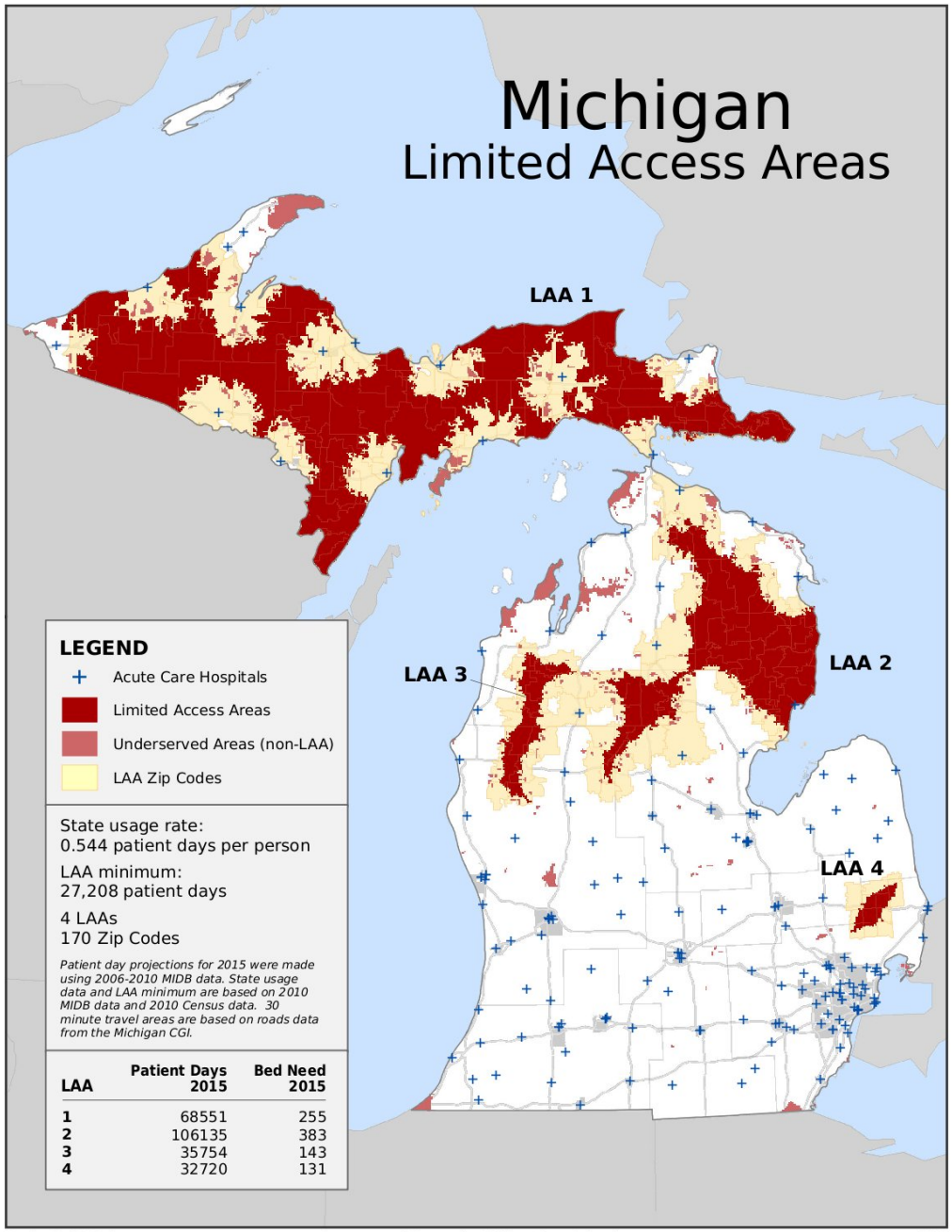
- Maps
  - Stand alone
    - Limited additional content or context
- Figures
  - Maps that are include in a publication, report, etc
    - Map is part of a larger document

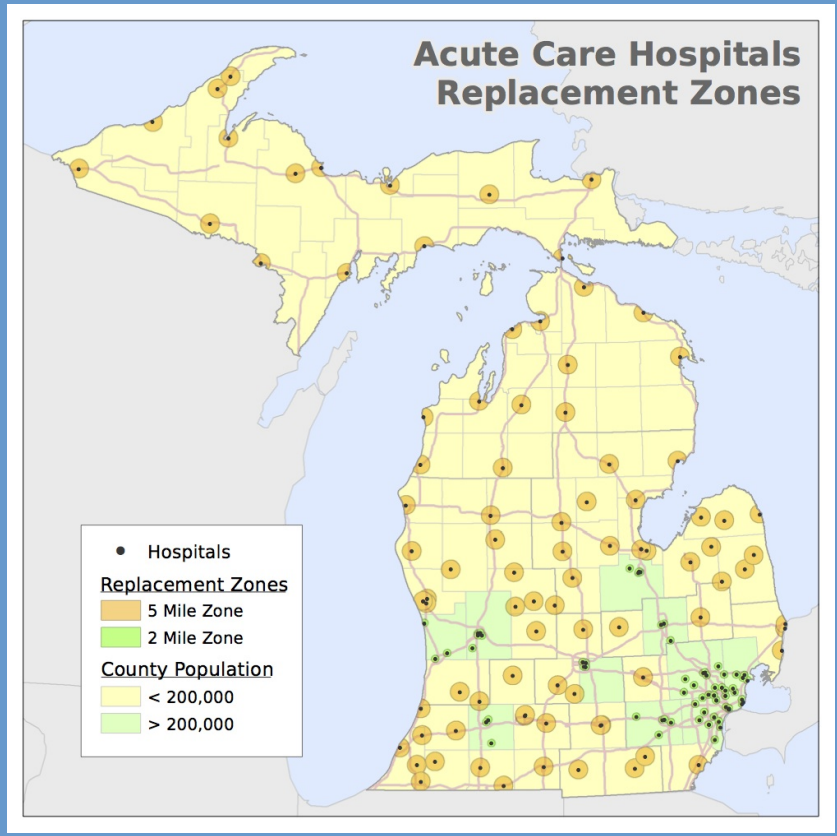
Figures/Maps

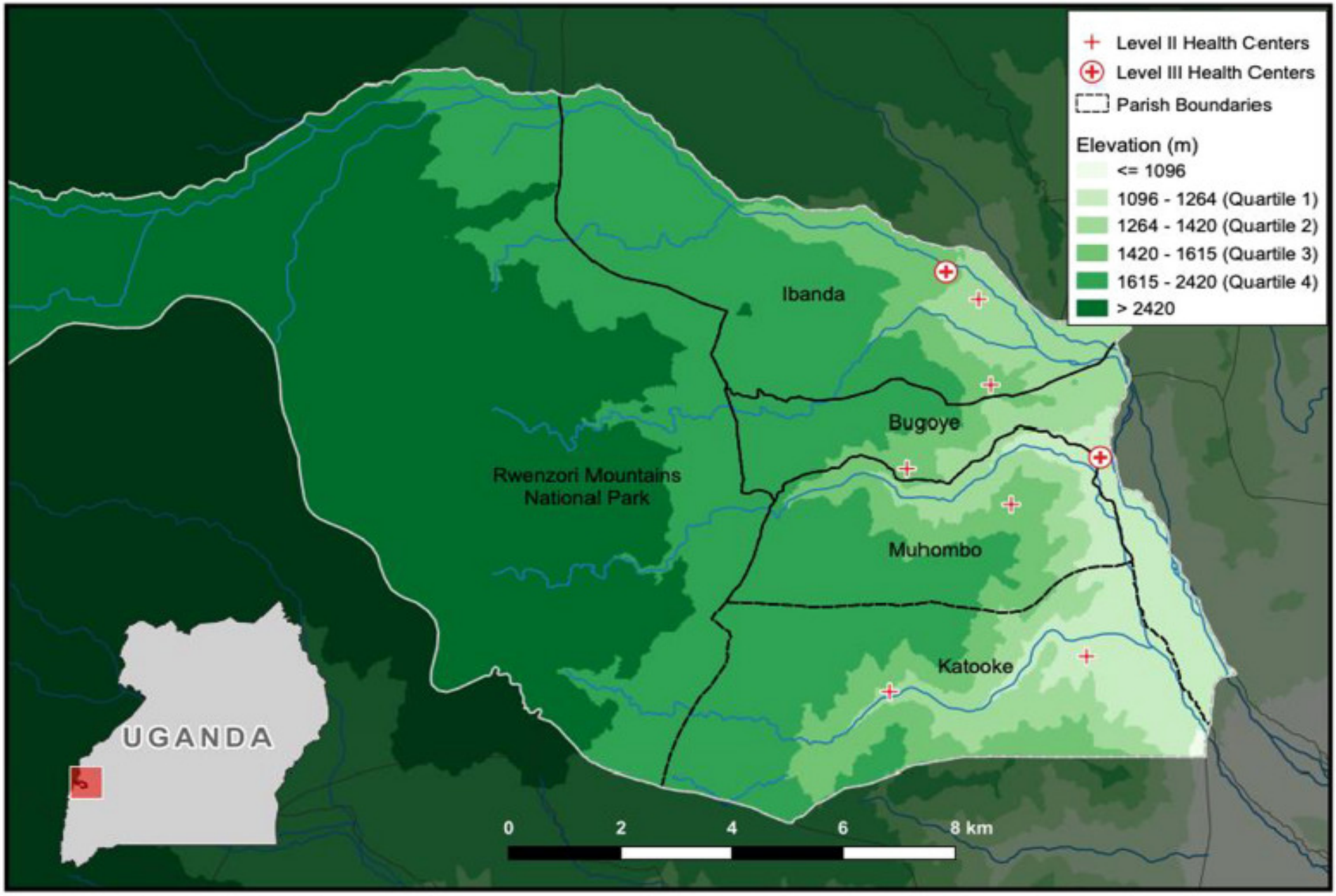
- Maps (stand alone) and Figure maps are very different
  - Level of detail included in a Figure map
    - *Probably* not necessary
      - North arrow (unless N anywhere but up)
      - Inset
    - *Definitely* not necessary
      - Data source
      - Author

Figures/Maps

- Maps (stand alone) and Figure maps are very different
  - Less is more in a Figure map
    - Thematic, not reference maps (generally)
      - Don't fill your maps with reference info or labeling, unless it is necessary, e.g., towns, counties, roads
    - Scale bar
      - If you used absolute "distance" in your analysis, then definitely yes

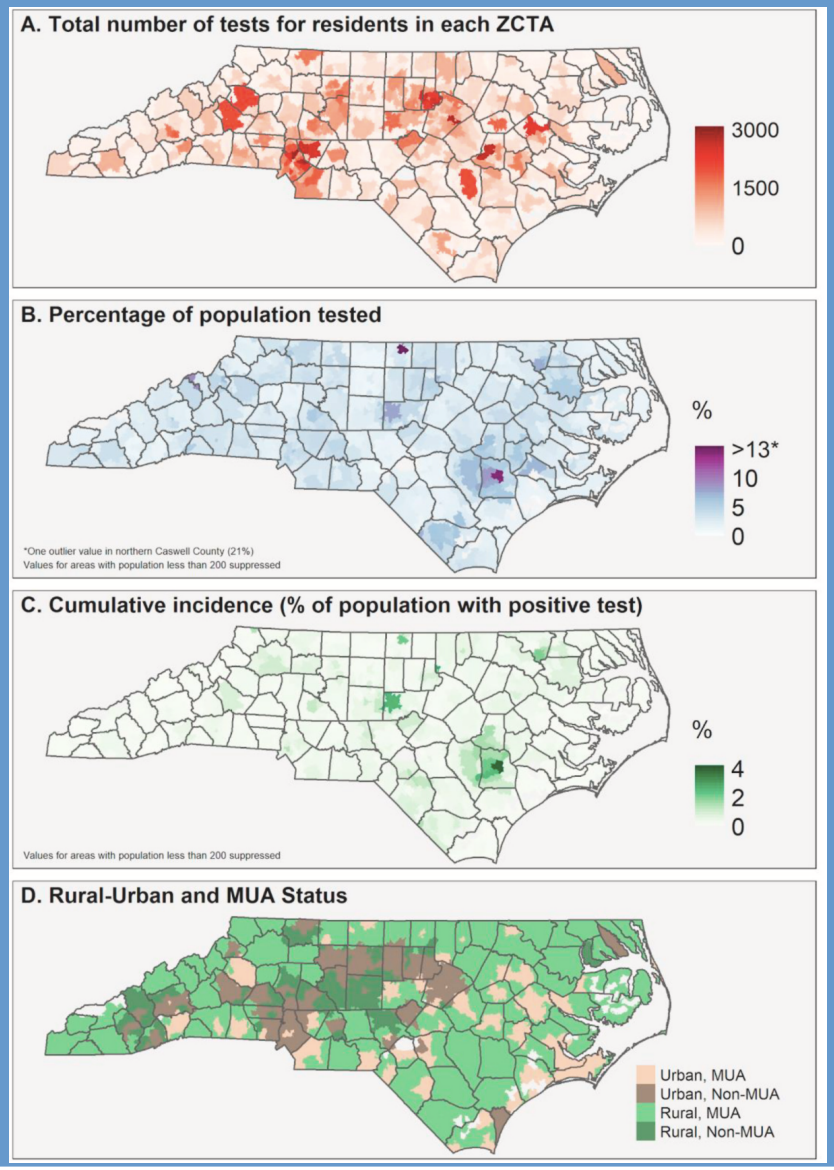

Michigan
Limited Access Areas
LAA 1
LAA 2
LAA 3
LAA 4
LEGEND
Acute Care Hospitals
Limited Access Areas
Underserved Areas (non-LAA)
LAA Zip Codes
State usage rate:
0.544 patient days per person
LAA minimum:
27,208 patient days
4 LAAs
170 Zip Codes
Patient day projections for 2015 were made using 2006-2010 MIDB data. State usage data and LAA minimum are based on 2010 MIDB data and 2010 Census data. 30 minute travel areas are based on roads data from the Michigan CGI.
LAA | Patient Days 2015 | Bed Need 2015
1 | 68551 | 255
2 | 106135 | 383
3 | 35754 | 143
4 | 32720 | 131
Map by: Paul L. Delamater
Department of Geography, Michigan State University
August, 2012


Acute Care Hospitals
Replacement Zones
Hospitals
Replacement Zones
5 Mile Zone
2 Mile Zone
County Population
< 200,000
> 200,000



Level II and Level III health centres.

Fig. 3. **Intersections between race and age in COVID-19 testing and positivity**. Fig. S2 in the Supplement extends the findings of Fig. 3A and B by showing median ages of individuals tested and individuals testing positive among sub-groups stratified by race-ethnicity and rural-urban status or gender.



Figures/Maps

- Should be created with the page layout and size of the journal in mind (when possible)
  - Text must be readable at the published page size
    - Text size (absolute minimum 7-8 pt)
    - No overlapping text
    - Stay away from things like auto-labels in QGIS
  - **Set the page size in QGIS!**

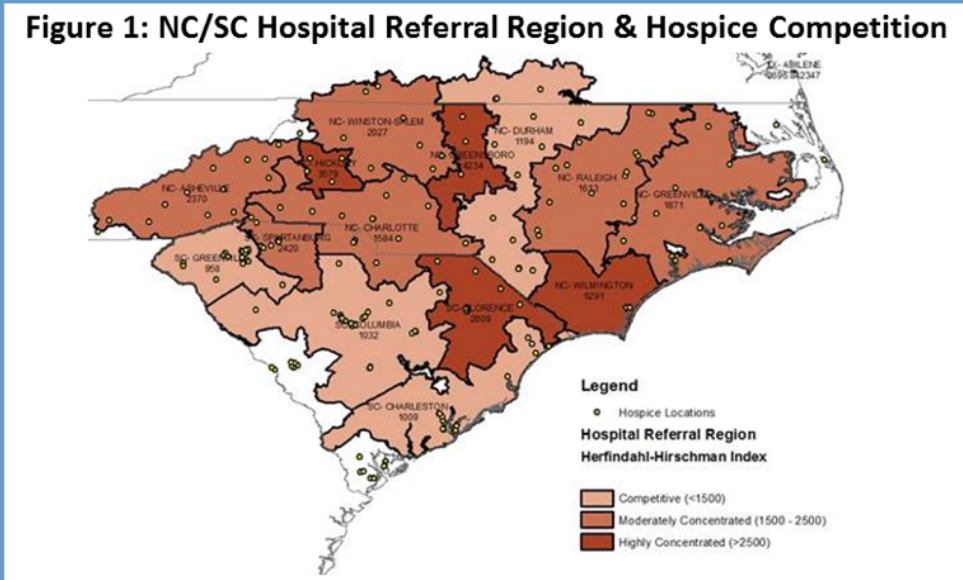
Map Design

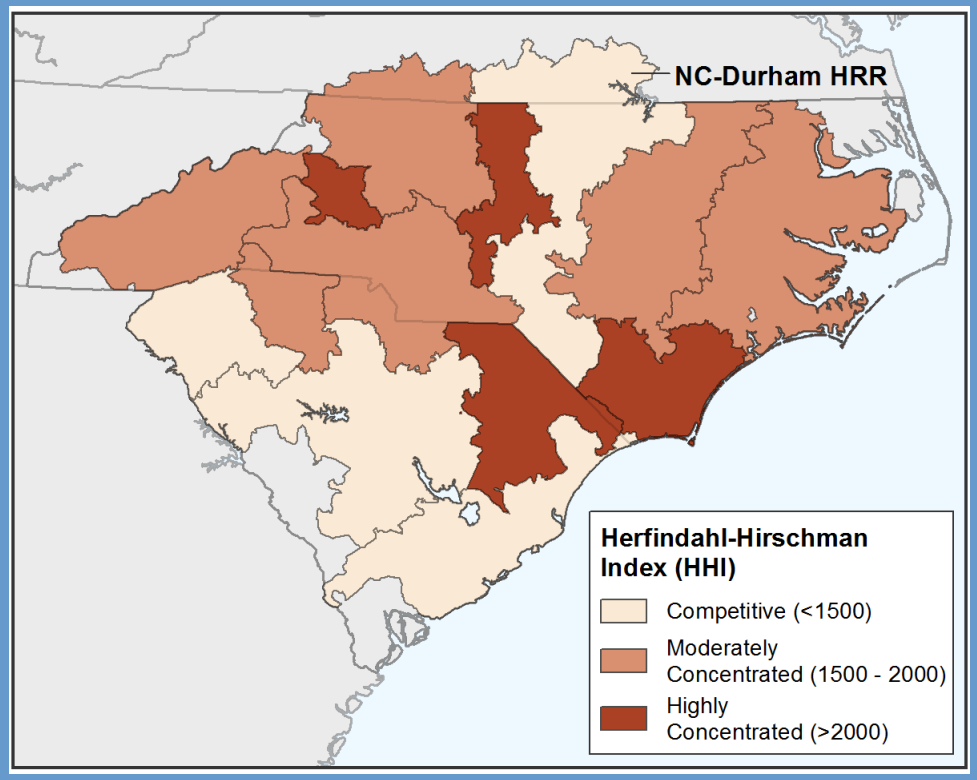
- It's the little things...
  - Background data
    - The dreaded "floating study area"
  - Legends!!
    - Default legends are awful
      - NO DEFAULT LAYER NAMES
      - Rounding of legend values
      - Units
      - Customize for your purpose (convert to graphics)

Map Design

- It's the little things...
  - Be proactive in your design choices
    - Colors
      - Please avoid...
        - Diverging color schemes when the "middle" value has no meaning
        - Multicolor schemes for a single variable, unless assists in interpretation
    - Line weights
    - Fonts
      - Arial or Times New Roman
    - Title
    - Consistency
      - Make one map, then use as a template

Map Design


Figure 1: NC/SC Hospital Referral Region & Hospice Competition
NC- WINSTON-SALEM
2027
NC- DURHAM
1194
NC- GREENSBORO
1234
NC- RALEIGH
1633
NC- GREENVILLE
1871
NC- ASHEVILLE
2370
NC- CHARLOTTE
1584
SC- SPARTANBURG
2420
SC- GREENVILLE
958
NC- WILMINGTON
5291
NC- FLORENCE
2609
SC- COLUMBIA
1032
SC- CHARLESTON
1009
Legend
Hospice Locations
Hospital Referral Region
Herfindahl-Hirschman Index
Competitive (<1500)
Moderately Concentrated (1500 - 2500)
Highly Concentrated (>2500)


NC-Durham HRR
Herfindahl-Hirschman Index (HHI)
Competitive (<1500)
Moderately Concentrated (1500 - 2000)
Highly Concentrated (>2000)

Keywords

- Selective content
- Symbol characteristics
- Visual variables
- Color ramps
  - Diverging schemes
- Map elements
- Design principles
  - Balance, hierarchy, figure-ground, legibility
- Figures vs Maps
  - Page size
- Little things
  - Floating study area, layer names, legend rounding

Wrap Up

- Next week
  - Query and Field Calculation; GIS Overlay, Distance, Buffer; Spatial Variation and Clustering
- Readings (next week)
  - C & M Chp 5